

Experiment 02C

Medical GPT Training with Custom BPE and Flash Attention

Medical text corpus | Custom biomedical BPE | Flash/SDPA Attention | Lightning H200 | 5 epochs

Report summary

This report documents the efficiency-oriented continuation of Experiment 02B. Experiment 02C keeps the custom biomedical BPE tokenizer and medical-domain training setup, while replacing standard attention with a flash/SDPA attention path. The objective is to preserve model quality while improving systems-level inference characteristics.

Dataset	Medical text corpus
Tokenizer	Custom biomedical BPE
Attention	Flash/SDPA attention
Model Scale	~194M parameters
Context Length	1024 tokens
Final Val Loss	2.1384
Final Perplexity	8.4861
Status	Efficiency-focused continuation of Experiment 02B

1. Objective

Experiment 02C evaluates whether the custom biomedical BPE model from Experiment 02B can maintain comparable language modeling quality when the attention implementation is changed from standard causal attention to a flash/SDPA attention path. The experiment is therefore focused on systems efficiency rather than tokenizer quality.

The main goal is to keep the medical-domain setting, tokenizer, model scale, and training schedule comparable while measuring the effect of an optimized attention implementation on training behavior and KV-cache inference benchmarks.

2. Experiment Design

Component	Configuration
Experiment role	Efficiency continuation after custom-tokenizer Experiment 02B
Tokenizer	Custom biomedical BPE
Vocabulary size	52,000
Attention type	Flash/SDPA attention
Dataset format	Pre-tokenized medical chunks
Hardware	Lightning AI H200
Epochs	5

3. Model and Training Configuration

Parameter	Value
Context length	1024
Embedding dimension	768
Attention heads	12
Transformer layers	12
Feedforward hidden dimension	3072
Dropout	0.1
QKV bias	False
Batch size	8
Gradient accumulation	8 steps
Effective batch size	64
Learning rate	3e-4

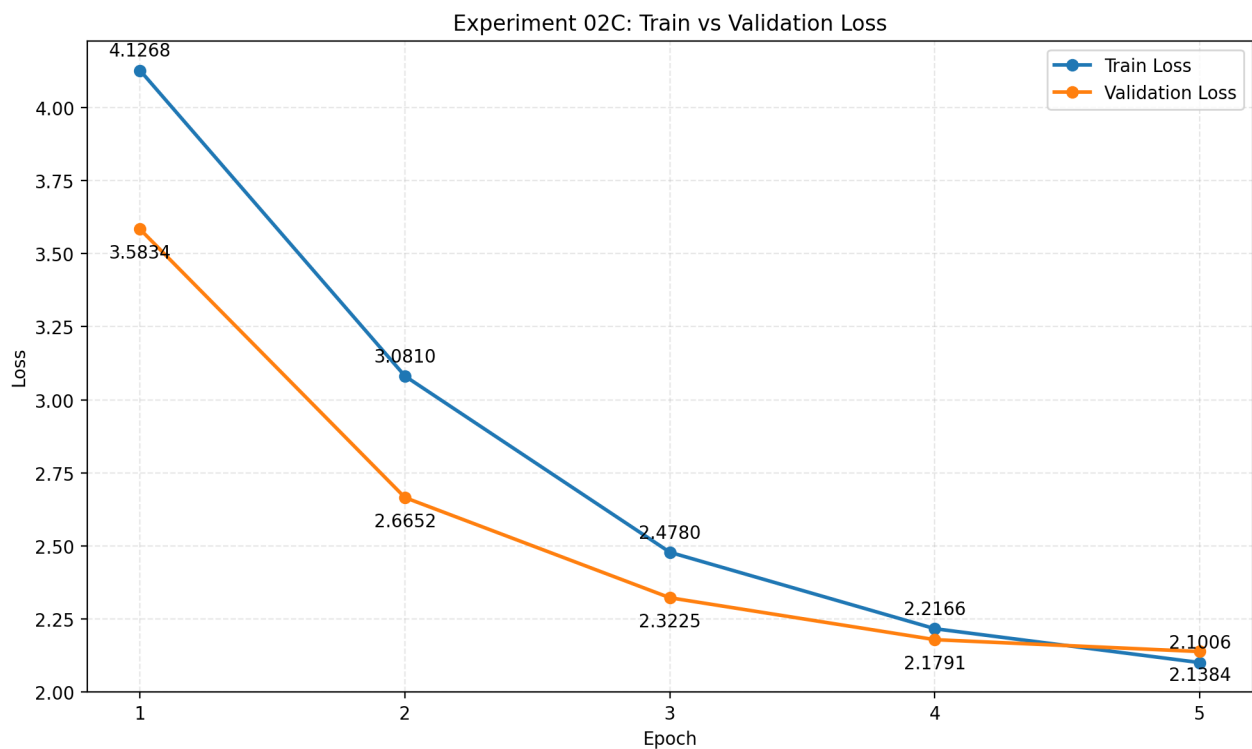
Parameter	Value
Minimum learning rate	1e-5
Checkpointing	best.pt, latest.pt, and epoch_N.pt

4. Training Results

Epoch	Optimizer Step	Train Loss	Val Loss	Perplexity	Learning Rate
1	420	4.126836	3.583400	35.9957	2.8382e-4
2	840	3.080969	2.665223	14.3712	2.1493e-4
3	1260	2.478034	2.322465	10.2008	1.2030e-4
4	1680	2.216577	2.179097	8.8383	4.0600e-5
5	2100	2.100648	2.138426	8.4861	1.0000e-5

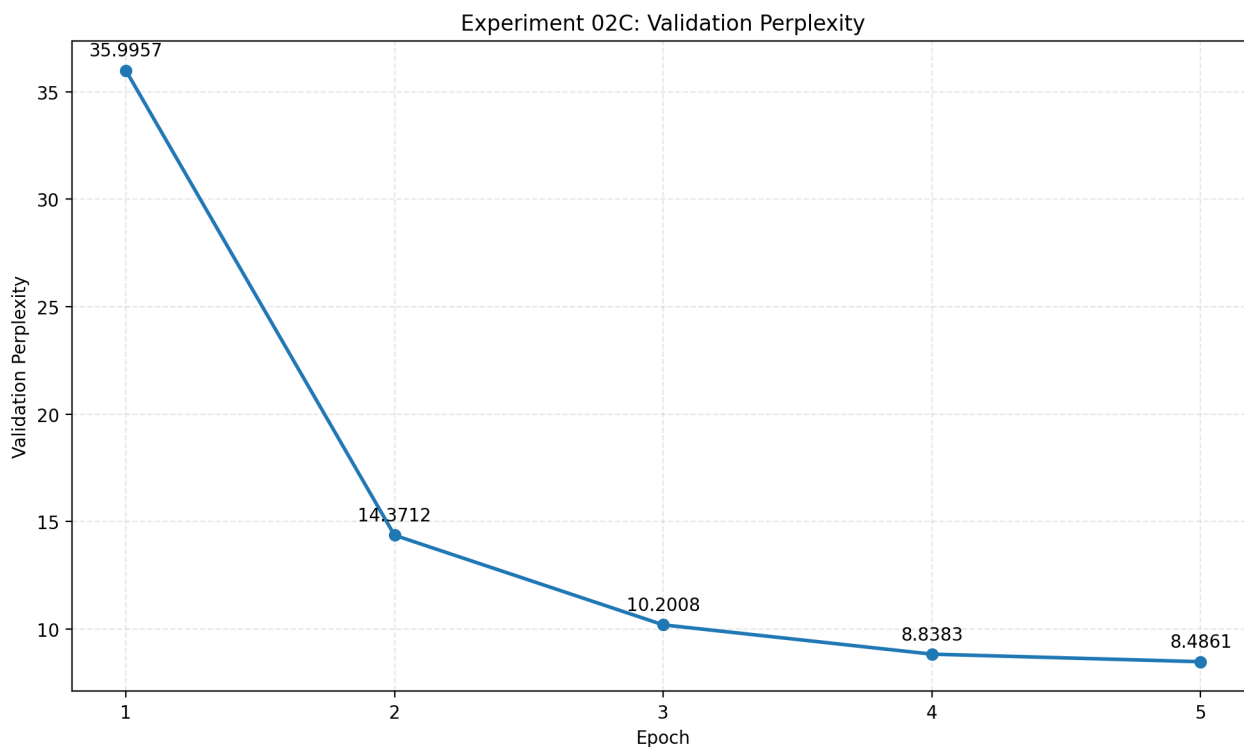
The model converged stably across all five epochs. Validation loss decreased from 3.5834 to 2.1384, while perplexity decreased from 35.9957 to 8.4861. The final result remains very close to the Experiment 02B standard-attention result, indicating that the optimized attention path preserved language modeling quality.

5. Loss Dynamics



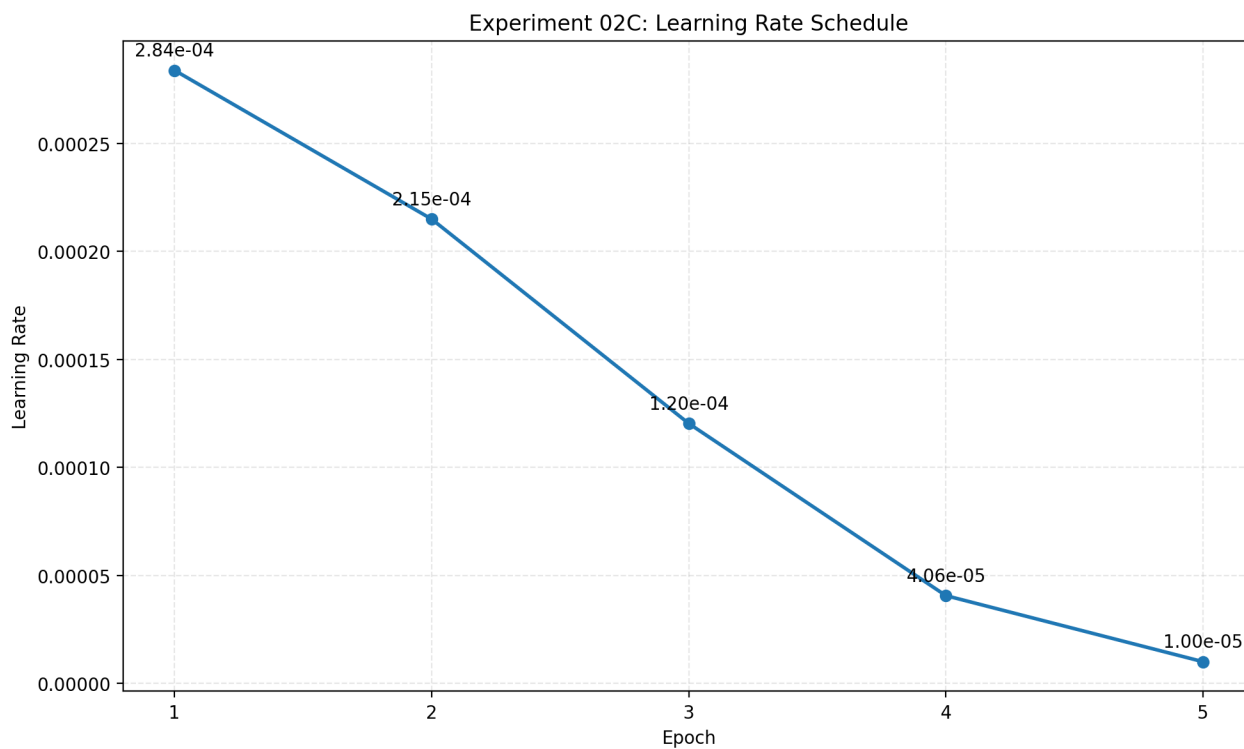
Training and validation loss both decrease across the experiment, indicating stable optimization.

6. Validation Perplexity



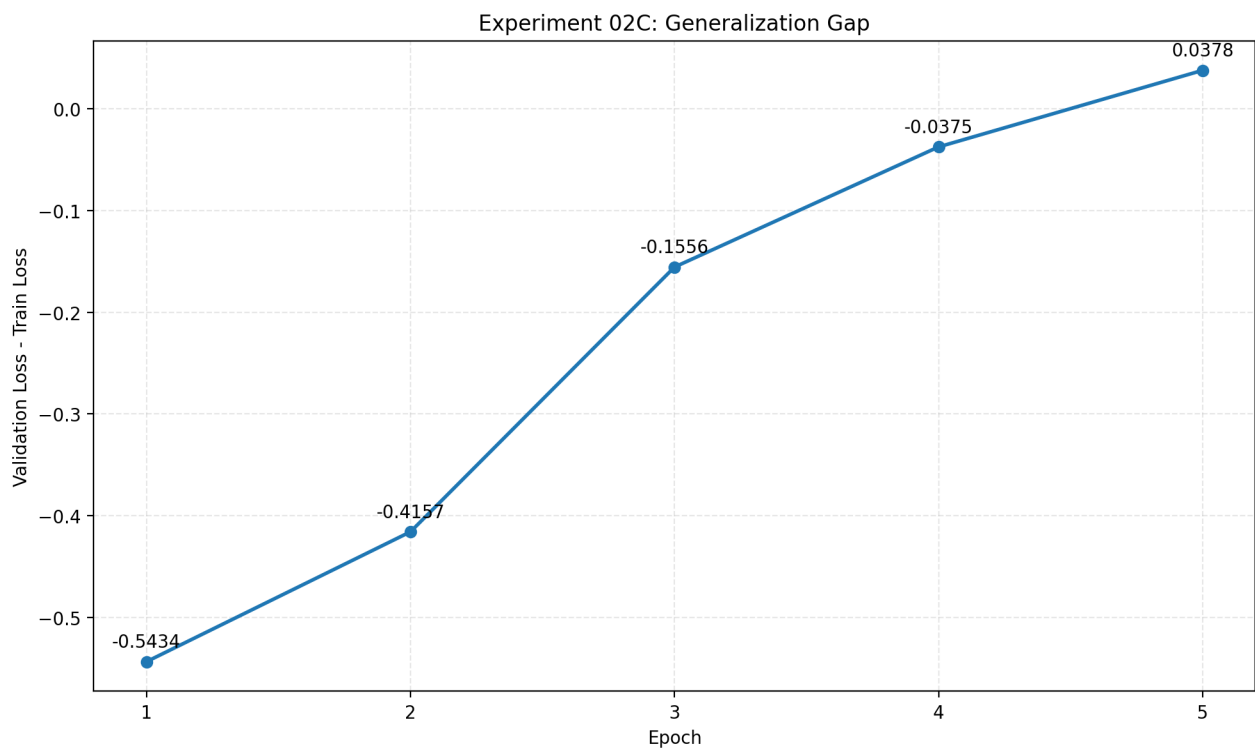
Validation perplexity improves sharply in the first two epochs and continues improving through epoch 5.

7. Learning Rate Schedule



The learning rate follows the planned warmup-plus-cosine decay behavior.

8. Generalization Gap



The gap remains small by the final epoch, showing controlled generalization behavior.

9. KV Cache Benchmark

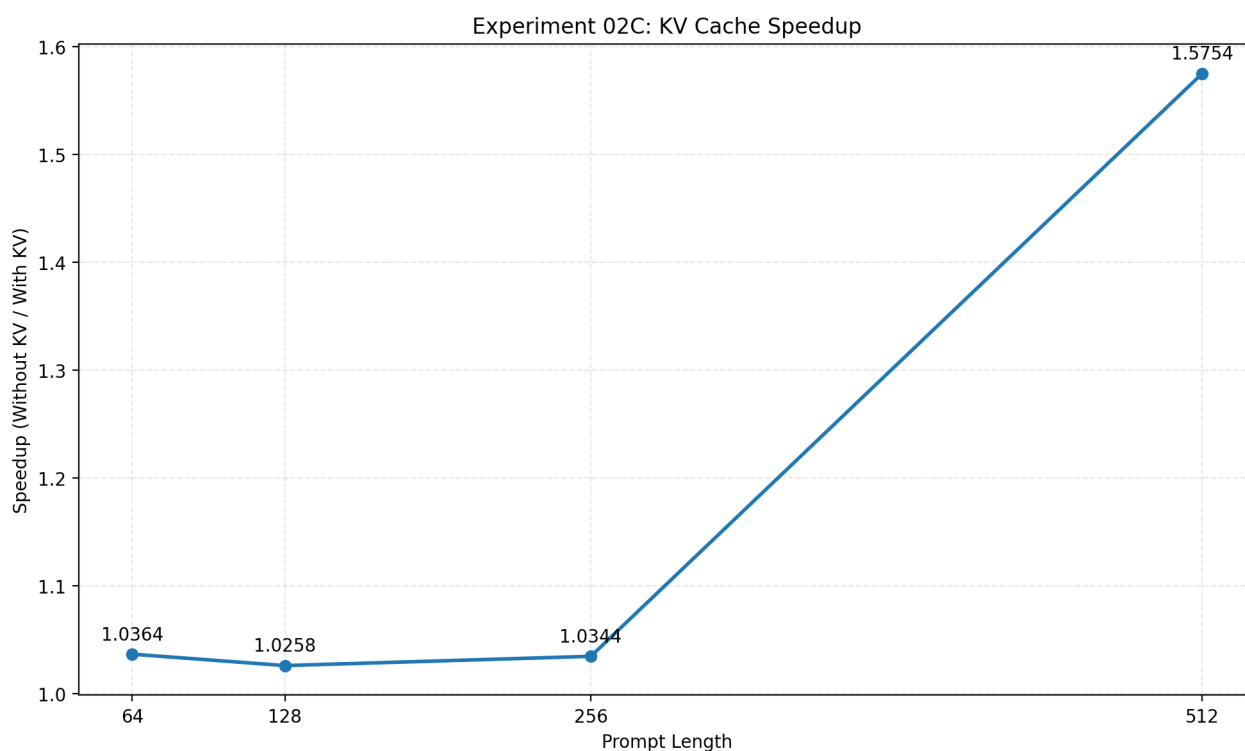
Inference benchmarking was conducted with and without KV-cache using prompt lengths of 64, 128, 256, and 512, with 64 generated tokens in each case. The benchmark reports latency, throughput, speedup, and peak memory behavior.

Prompt	Max New	Latency w/o KV	Latency w/ KV	Speedup
64	64	0.290073	0.279894	1.0364x
128	64	0.289977	0.282696	1.0258x
256	64	0.296323	0.286467	1.0344x
512	64	0.441901	0.280507	1.5754x

Prompt	TPS w/o KV	TPS w/ KV	Memory w/o KV	Memory w/ KV
64	220.63	228.66	823.13 MB	795.98 MB
128	220.71	226.39	848.32 MB	829.60 MB
256	216.20	223.45	900.70 MB	867.82 MB
512	144.83	228.17	1001.83 MB	975.28 MB

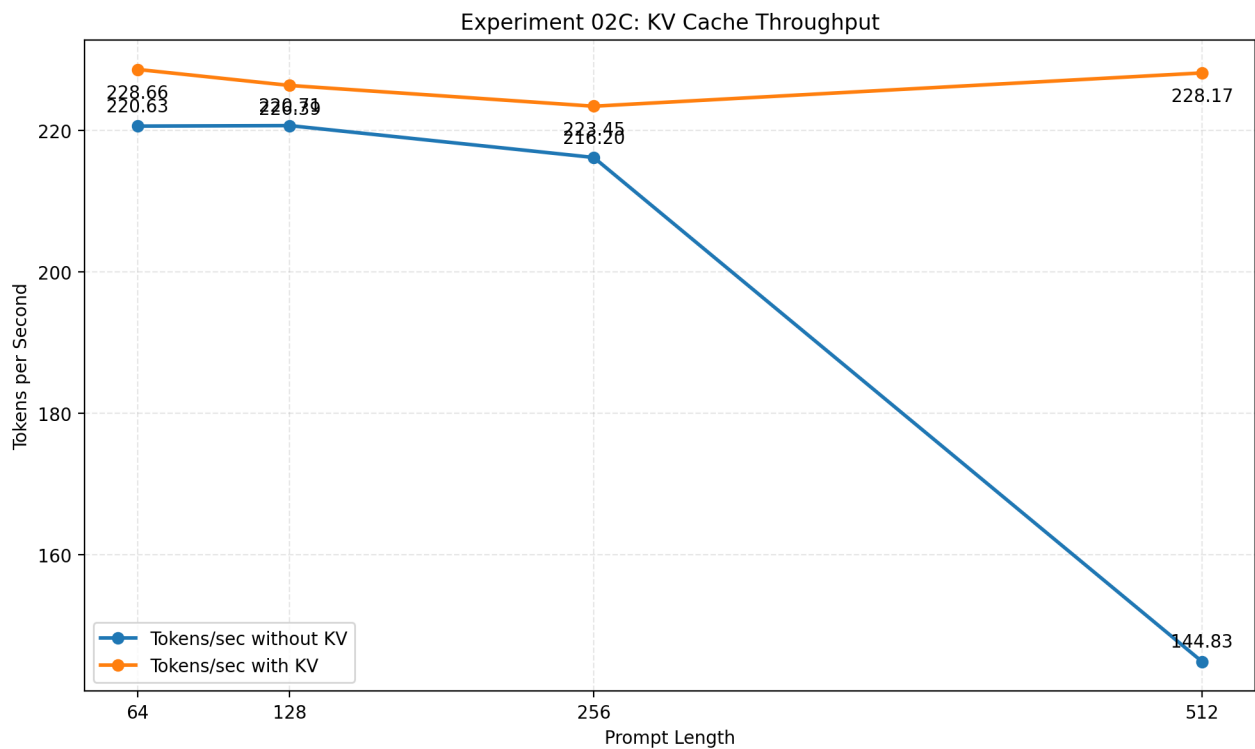
The strongest KV-cache benefit occurs at prompt length 512, where latency decreases from 0.441901 seconds to 0.280507 seconds and throughput improves from 144.83 tokens/sec to 228.17 tokens/sec.

10. KV Cache Speedup



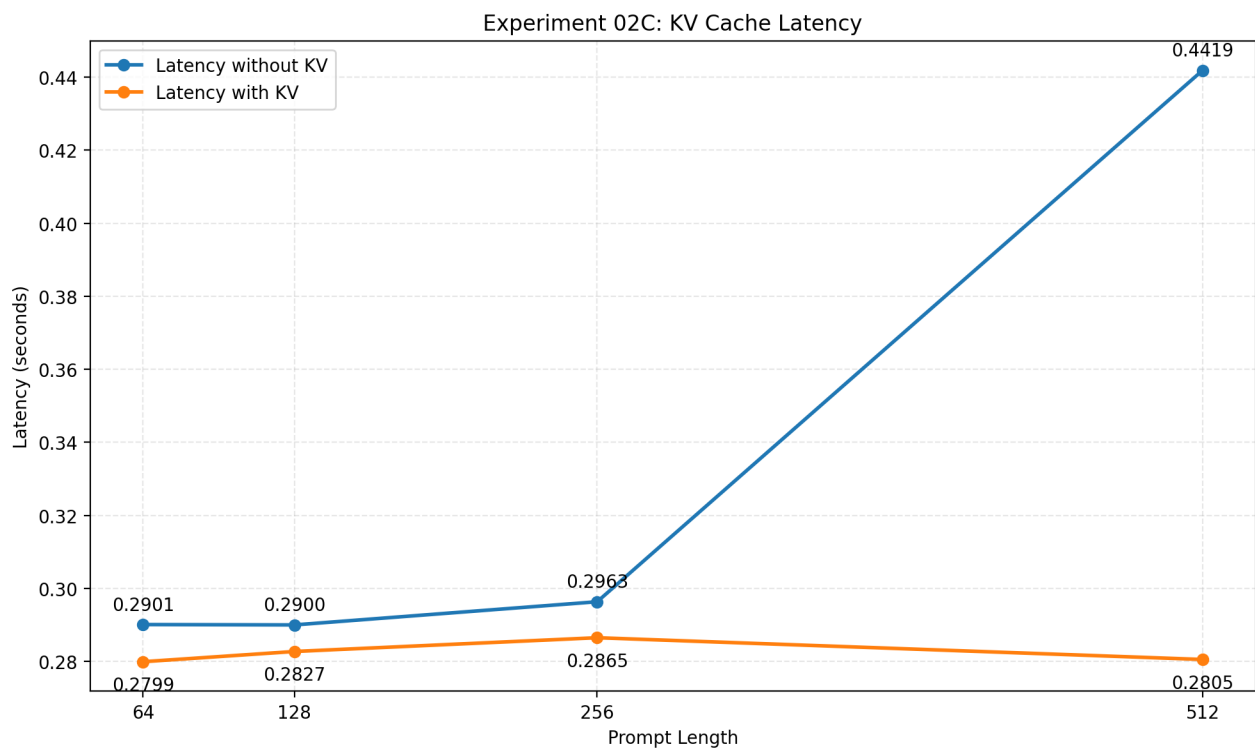
KV-cache speedup is strongest for the longest prompt length.

11. KV Cache Throughput



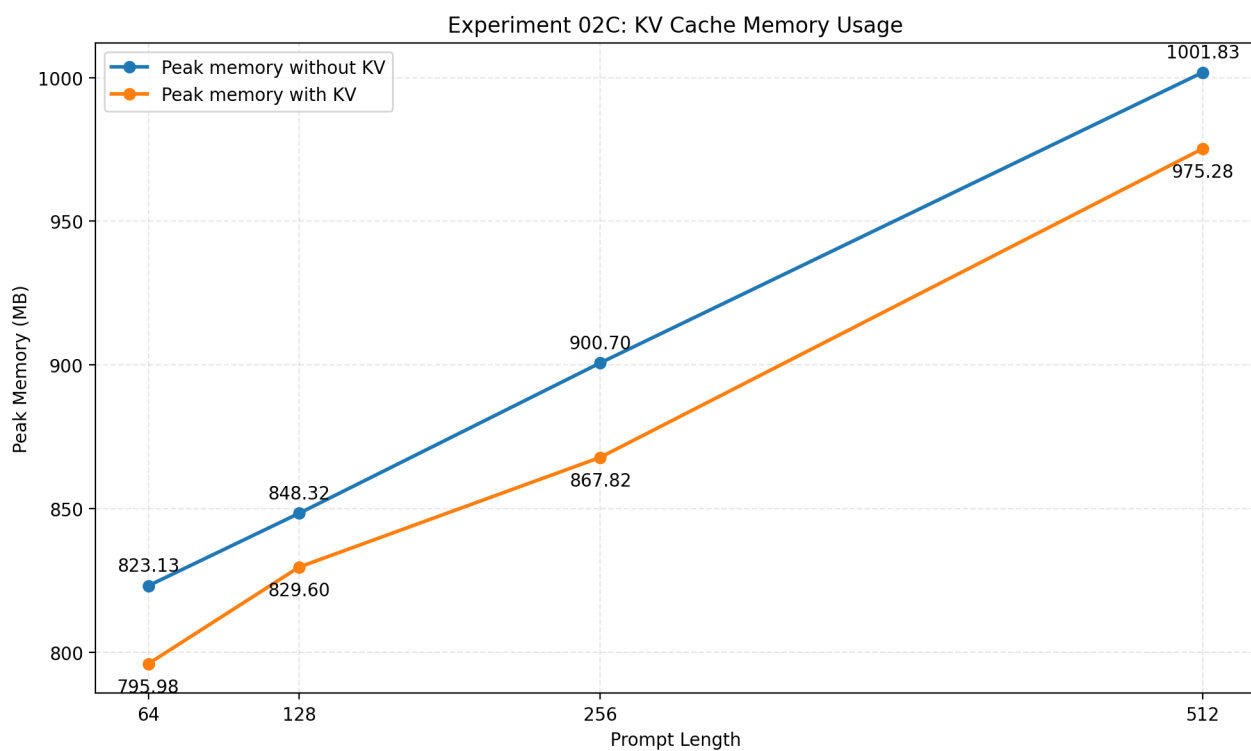
Cached decoding consistently improves tokens-per-second across the tested prompt lengths.

12. KV Cache Latency



KV-cache reduces latency most clearly at prompt length 512.

13. KV Cache Memory Usage



Peak memory remains controlled while cached decoding improves throughput.

14. Comparison with Experiment 02B

Metric	Experiment 02B	Experiment 02C
Tokenizer	Custom biomedical BPE	Custom biomedical BPE
Attention	Standard causal attention	Flash/SDPA attention
Final train loss	2.0879	2.1006
Final validation loss	2.1301	2.1384
Final perplexity	8.4161	8.4861
Best KV speedup	1.4896x	1.5754x
Best cached throughput	210.76 tok/s	228.17 tok/s

Experiment 02C maintained nearly the same validation quality as Experiment 02B while improving the best KV-cache speedup and cached throughput. This supports the conclusion that the flash/SDPA attention path is a useful implementation-level optimization for the project.

15. Interpretation

The small difference between the Experiment 02B and 02C validation metrics indicates that the optimized attention path did not materially degrade model quality. The main benefit of Experiment 02C is therefore not lower loss, but better systems behavior during inference.

The KV-cache benchmark confirms the expected pattern: the benefit of cached decoding increases at longer prompt lengths because the model avoids repeatedly recomputing attention states for previously processed tokens.

16. Future Work

Future work can extend this line of experimentation by scaling training duration, increasing corpus size, and testing whether the same efficiency behavior remains stable under larger context windows. Additional studies can also compare generation quality, domain terminology handling, and throughput under different decoding settings.

17. Conclusion

Experiment 02C successfully extended the custom-tokenizer medical GPT setup with flash/SDPA attention. The model achieved a final validation loss of 2.1384 and a final perplexity of 8.4861, closely matching the standard-attention baseline from Experiment 02B. At the same time, the KV-cache benchmark showed stronger long-prompt inference behavior, with the best speedup reaching 1.5754x at prompt length 512.

Overall, Experiment 02C demonstrates that the project can preserve medical-domain modeling quality while improving implementation-level efficiency.